



Project Proposal

Group Name: Hidden Fire

Submitted To:

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Course Title:

Mobile Application Development

Course code:

CS-610

Department Of Computer Science

Section: B

UNIVERSITY OF LAYYAH

Project Title:

MoneyFlow AI (Smart Personal Finance Management Application)

1. Introduction

MoneyFlow AI is a mobile-based financial management application designed to help users track their income, expenses, budgets, and financial habits in an efficient and smart way. The application provides real-time financial insights, category-based tracking, and visual reports to improve personal financial decision-making.

2. Background of the Study

In modern life, managing personal finances manually is difficult due to frequent transactions and lack of tracking tools. Many individuals struggle with overspending and poor budgeting. Existing solutions often lack simplicity or smart insights. Therefore, a user-friendly and intelligent mobile application is required to solve these issues.

3. Problem Statement

Most individuals face the following problems:

- Difficulty in tracking daily expenses
- Lack of budget control
- No proper financial analysis tools
- Manual and unorganized record keeping
- No alerts for overspending

4. Objectives of the Project

- To develop a smart expense tracking system
- To help users manage income and expenses efficiently
- To provide budget monitoring and alerts
- To visualize financial data using charts
- To improve financial awareness among users

5. Proposed Solution

MoneyFlow AI solves financial management problems by providing:

- Automated transaction tracking
- Budget management system
- Smart alerts and notifications
- Data visualization (charts & reports)
- Secure user-based financial storage

6. Features of the Application

Splash Screen

- App launch par animated splash screen
- App logo and name display
- Short loading time before main app

Income Management

- Add income
- Edit income
- Delete income
- View total income

Expense Management

- Add expense
- Edit expense
- Delete expense
- View total expenses

Budget Management

- Set monthly budget
- Automatically calculate remaining budget
- Show budget usage percentage
- Track daily expenses

Transaction History

- View complete transaction list
- Date-wise transaction display
- Add short notes with transactions
- Separate view for income and expenses

Category System

- Add categories to expenses (Food, Transport, Bills, etc.)
- Mark important expenses
- Filter transactions by category

Alert System

- Warning alert at 80% budget usage
- Full budget alert at 100%
- Daily overspending notifications

Report System

- View in-app financial summary report
- Visual summary of income and expenses
- Take screenshot for sharing/exporting

User Authentication System

- User signup
- User login
- Secure session management
- User-specific data storage

Profile Screen

- View user profile (name, email, profile picture)
- Edit profile information
- Update profile picture
- Account details management
- logout

Data Visualization

- Pie chart for category-wise spending
- Bar chart for monthly expense comparison
- Budget progress bar
- Visual expense breakdown

Security System

- PIN lock for app
- Secure access to financial data
- Authentication-based protection

Search & Filter System

- Search transactions by name or note
- Filter data by date, category, or type (income/expense)
- Quickly find specific records

Dashboard Summary

- Show daily, weekly, and monthly overview
- Display total income, expenses, and balance on one screen
- Quick financial snapshot for user convenience

Firebase Structure

Authentication

- Firebase Authentication (Email & Password)
- User signup and login system
- Each user has a unique UID

Users Collection

Collection: users

- name
- email
- createdAt
- uid
- profile Image

Transactions Collection

Collection: transactions

- userId (link to user)
- title
- amount
- type (income / expense)
- category
- date
- note
- isRecurring
- createdAt (for sorting)

Budget Collection

Collection: budgets

- userId
- monthlyBudget
- month
- year

Remaining budget is calculated in the app from transactions (not stored in Firebase)

7. Target Users

- Students who want to manage their daily expenses and pocket money
- Working professionals who need to track income, expenses, and savings
- Small business owners for basic financial tracking
- Individuals managing personal or household budgets
- Anyone looking to improve financial awareness and spending habits

8. Tools and Technologies

- Android Studio
- Java / Kotlin
- Firebase Authentication
- Cloud Firestore Database
- MPAndroidChart (for graphs)
- Material Design UI (M3)

9. System Requirements

Hardware Requirements:

- Android Mobile Device (4GB RAM recommended)
- Minimum 64GB storage

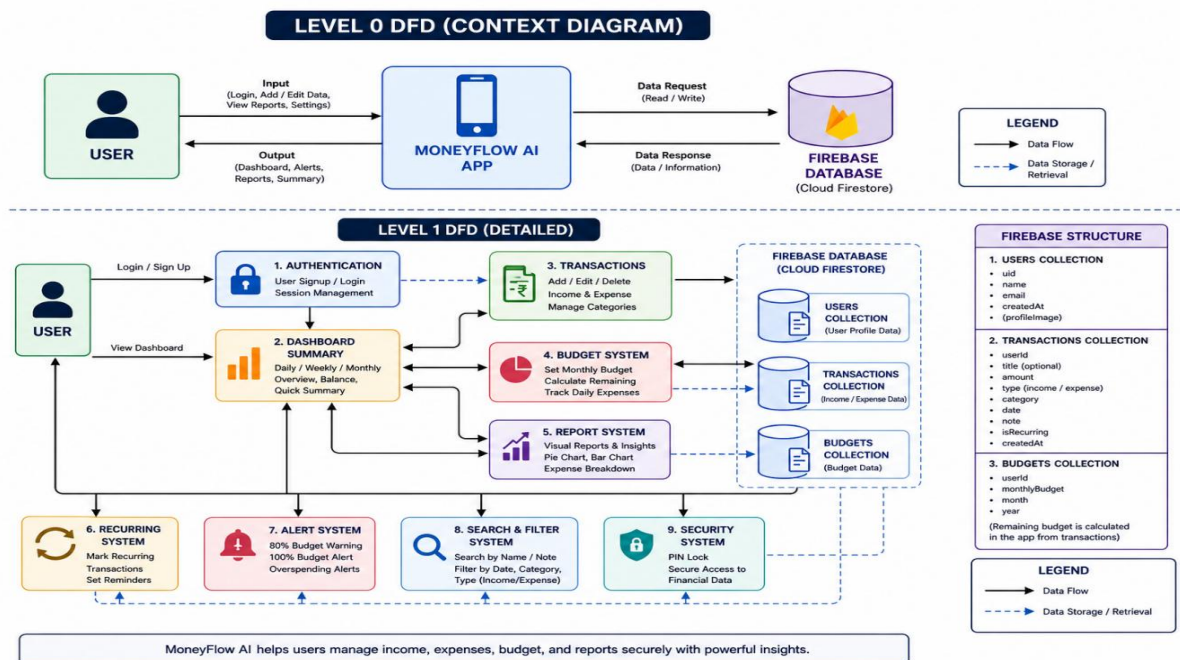
Software Requirements:

- Android 8.0 or above
- Internet connection
- Firebase backend services

10. Methodology / Development Process

- Requirement Analysis
- UI/UX Design
- Database Design (Firebase)
- Application Development
- Testing & Debugging
- Deployment

11. System Architecture / Data Flow Diagram (DFD)

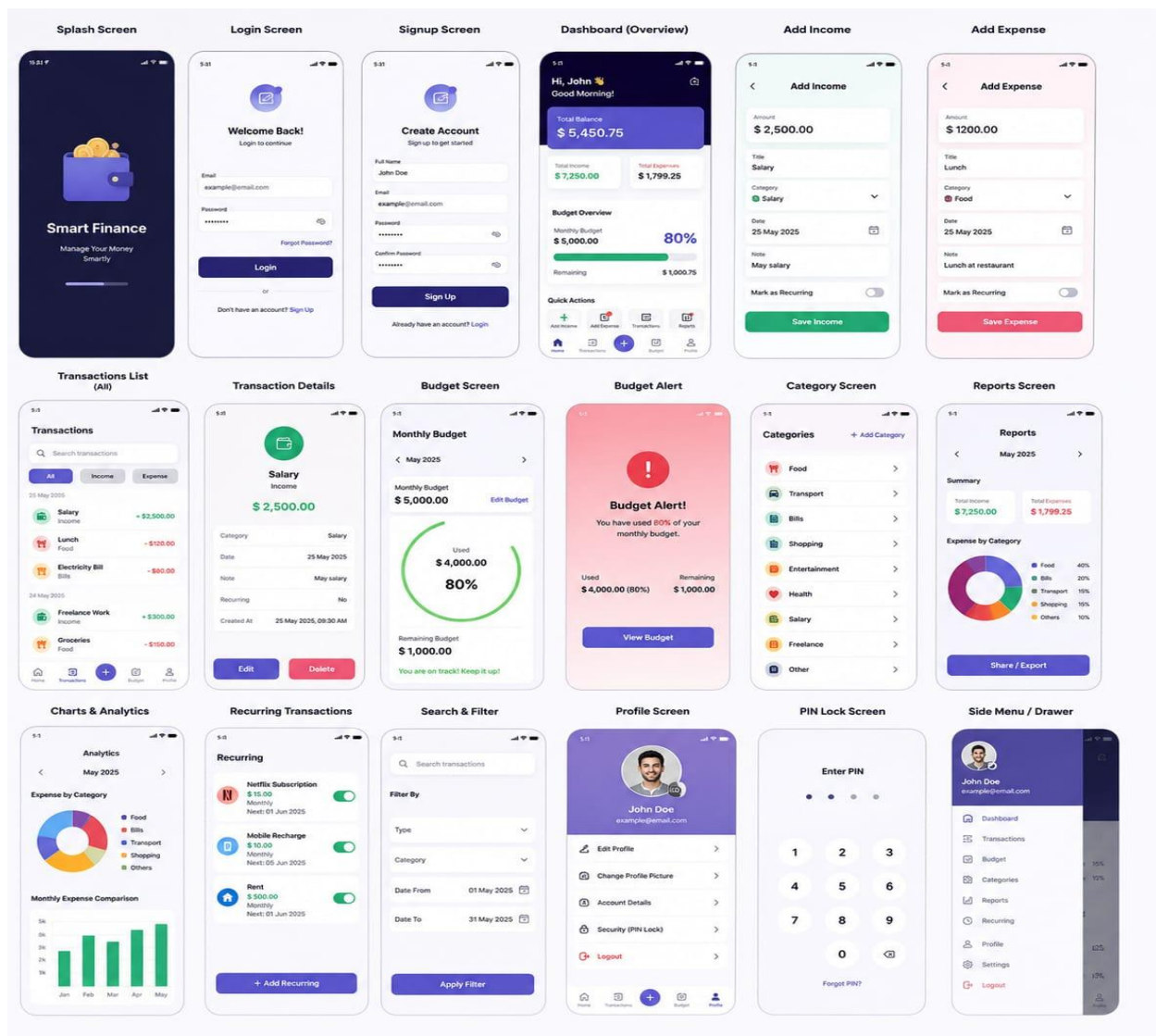


12. Testing Strategy

- Unit Testing (individual features)
- Integration Testing (Firebase connection)
- UI Testing (user interface validation)
- Performance Testing (app speed and response)

- User Acceptance Testing

13. User Interface Screens / Mockups



14. Expected Results / Output

- Fully functional expense tracking app
- Real-time financial updates
- Accurate budget calculations
- Visual financial reports

- Secure user data storage

15. Future Enhancements

- AI-based spending predictions
- Bank account integration
- Cloud backup system
- Multi-language support
- Advanced financial recommendations

16. Conclusion

MoneyFlow AI is an efficient and smart financial management application designed to help users control their finances effectively. It simplifies budgeting, enhances financial awareness, and provides meaningful insights through visual data representation.

17. References

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<https://play.google.com/store/search?q=money%20manager%20expense%20tracker&c=apps>

[5] Wallet App – <https://budgetbakers.com/wallet>

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[7] MPAndroidChart – <https://github.com/PhilJay/MPAndroidChart>

[8] Material Design – <https://m3.material.io/>

[9] GeeksforGeeks Expense Tracker App – <https://www.geeksforgeeks.org/android-expense-tracker-app-using-firebase/>

★ End of Proposal ★